A Time-Varying Analysis of the Expectations–Policy Relationship in Sweden, 2010–2024

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Abstract

This paper examines how the Riksbank's responsiveness to inflation expectations evolved over 2010–2024, addressing the research question: *How has the Riksbank's responsiveness to inflation expectations evolved over time, and does the influence of expectations on Swedish monetary policy vary across regimes such as the low-inflation period, negative interest rate era, and the 2021–2023 inflation surge?* Using monthly Swedish data and a time-varying Taylor-type rule estimated in a state-space framework via the Kalman filter, the policy coefficients on inflation, expectations, and slack are allowed to drift gradually over time. The expectations coefficient becomes firmly positive after 2012 and remains economically important in all regimes, with its strongest influence during the negative-rate and QE period. The inflation coefficient is weak and unstable for most of the 2010s but turns decisively positive during the post-2021 inflation surge, while slack plays only a limited role throughout the entire sample. The time-varying rule closely matches the observed repo rate, indicating that shifting reaction coefficients are key to capturing Swedish monetary policy behavior. Overall, the results reveal systematic adjustments that would be obscured under a constant-parameter rule.

Introduction

Inflation expectations play a central role in modern monetary policy. For an inflation-targeting central bank, credibility depends on whether expectations remain anchored and on how strongly policymakers react when expectations deviate from target. Swedish survey evidence indicates that inflation expectations are informative and closely linked to inflation dynamics (Jonsson and Österholm, 2009). However, monetary policy rules do not need to be constant. The responsiveness of the policy rate to inflation, real activity, or expectations may vary across regimes as economic conditions change over time. This is especially relevant for Sweden, where the time span between 2010–2024 featured prolonged low inflation, negative interest rates, large-scale asset purchases, as well as a sharp inflation surge during 2021–2023. A single constant-parameter Taylor rule estimated over this entire period would obscure regime-specific changes in how policymakers weigh inflation, slack, and expectations. This motivates a time-varying approach. Existing Swedish studies typically use constant-parameter Taylor rules or allow only for discrete breaks (Jonsson et al., 2017; Lundmark, 2023). Such approaches cannot capture gradual structural shifts, since constant-coefficient rules impose a fixed policy reaction over the entire sample, meaning they can only accommodate abrupt changes and will average over slow-moving behavioral adjustments, masking shifts in how the Riksbank responds to inflation, slack, or expectations. By contrast, the international literature documents substantial time variation in monetary policy behavior using time-varying parameter (TVP) models (Cogley and Sargent, 2001a, 2005a; Primiceri, 2005; Yüksel, 2013; Anderl, 2024; Brault, 2025). Preliminary Swedish evidence also suggests evolving expectations dynamics: Engdahl (2024) finds that inflation expectations Granger-cause the policy rate more consistently than the reverse direction in a reduced-form VAR for 2010–2024. Motivated by these insights, this paper estimates a time-varying Taylor-type reaction function in a state-space form using the Kalman filter. The framework focuses on drifting coefficients rather than full multivariate time-varying volatility and allows the Riksbank's responses to inflation, expectations, and slack to evolve smoothly over time. Using monthly data for 2010–2024, I estimate the time paths of the key reaction coefficients, with particular attention to the evolving weight on inflation expectations and how this differs across policy regimes. I aim to make a twofold contribution. First, this is, to my knowledge, the first empirical estimate of a Swedish Taylor rule with an explicitly time-varying expectations coefficient, complementing the constant-parameter literature (Jonsson et al., 2017; Lundmark, 2023). Second, the results offer new evidence on how Swedish monetary policy adapted during periods of below-target inflation, unconventional policy, and the post-pandemic inflation surge, adding a time-varying dimension to the analysis of the Riksbank's behavior.

1 Methodology

This section outlines the time-varying Taylor rule framework used to analyse how the Riksbank's responsiveness to inflation, inflation expectations, and economic slack evolved during 2010–2024. The empirical approach follows the large literature on time-varying parameter (TVP) models in macroeconomics, where policy coefficients are allowed to drift gradually over time in response to changes in credibility, uncertainty, or preferences (Cogley and Sargent, 2001b, 2005b; Primiceri, 2005). While these studies typically estimate multivariate TVP-VARs using Bayesian MCMC methods, I apply a more compact and transparent state-space specification estimated using the Kalman filter.

1.1 Time-Varying Taylor Rule

Monetary policy is modeled using a forward-looking Taylor-type reaction function with time-varying parameters:

$$i_t = \rho i_{t-1} + \alpha_t(\pi_t - \pi^*) + \beta_t(\exp_t - \pi^*) + \gamma_t \, \mathrm{gap}_t + \varepsilon_t, \tag{1}$$

where i_t is the policy rate, π_t denotes CPIF inflation (year-on-year), with $\pi^* = 2$ denoting the inflation target, $\exp_t$ is one-year-ahead inflation expectations, and gap_t is a measure of economic slack. In the empirical application below, the expectations term is based on CPI rather than CPIF, reflecting data availability; the two series track each other closely at the relevant horizon.

The coefficients α_t, β_t, and γ_t are allowed to evolve over time, capturing gradual changes in policy behavior without imposing discrete regime shifts. The residual is assumed Gaussian with constant variance,

$$\varepsilon_t \sim \mathcal{N}(0, \sigma^2),$$

and stochastic volatility is left for future work.

1.2 State-Space Representation

Equation (1) can be written in state-space form. The measurement equation relates the observable policy rate to the latent coefficient vector:

$$i_t = \rho i_{t-1} + X_t' \theta_t + \varepsilon_t, \tag{2}$$

where $X_t = (x_{1,t}, x_{2,t}, x_{3,t})'$ with $x_{1,t} = (\pi_t - \pi^*)$, $x_{2,t} = (\exp_t - \pi^*)$, and $x_{3,t} = \mathrm{gap}_t$, and where $\theta_t = (\alpha_t, \beta_t, \gamma_t)'$.

The time-variation in the coefficients is captured by a random-walk transition equation:

$$\theta_t = \theta_{t-1} + u_t, \qquad u_t \sim \mathcal{N}(0, Q),$$

where Q is a diagonal variance matrix. This specification follows the coefficient-drift component of the models in Cogley and Sargent (2005b) and Primiceri (2005), without modeling time variation in shock variances or contemporaneous relations.

Before estimation, the regressors in X_t are standardized to have zero mean and unit variance. Scaling is recommended in state-space maximum-likelihood estimation because it improves numerical conditioning of the likelihood function and stabilises the optimisation of the variance parameters. As noted by Harvey (1989) and Durbin and Koopman (2012), large differences in the magnitudes of the explanatory variables can lead to ill-conditioned Hessians and poor convergence of the Kalman filter likelihood. Similar standardization procedures are commonly applied in empirical TVP models, including the multivariate frameworks of Koop and Korobilis (2010, 2013). Since coefficient paths are recovered on the scaled units, the economic interpretation remains unchanged.

1.3 Estimation via the Kalman Filter

The state-space model is estimated using maximum likelihood combined with the Kalman filter and a fixed-interval smoother. Conditional on the variance parameters (σ^2, Q), the Kalman filter provides filtered estimates of the coefficient vector based on the available information at time t, while the smoother recovers full-sample trajectories of the latent states. The variance parameters are estimated by maximizing the Gaussian likelihood of the state-space system (Durbin and Koopman, 2012; Harvey, 1989). Under Gaussian assumptions, the filtering step can be interpreted as exact Bayesian updating of the latent states conditional on fixed hyper-parameters, although the overall estimation procedure remains likelihood-based rather than fully Bayesian.

1.4 Interpretation

The time paths of α_t, β_t, and γ_t reveal how the monetary policy rule evolves over time:

- β_t measures the response to deviations of inflation expectations from target.
- α_t measures the response to realized inflation.
- γ_t measures the response to economic slack.

Changes in these coefficients reveal shifts in the expectations policy relationship across different monetary policy regimes. The random-walk specification allows for gradual drift, as stated in the TVP literature (Primiceri, 2005; Canova and Gambetti, 2009).

Data

The empirical analysis of this paper uses monthly Swedish data for the period 2010–2024. Beginning the sample in 2010 ensures consistent availability of survey-based inflation expectations, while covering the full sequence of monetary policy regimes with low inflation, negative interest rates, quantitative easing, as well as the post-2021 inflation surge. The baseline time-varying policy rule includes four observed series:

- **Repo rate** (i_t): the Riksbank's policy rate, end-of-month, obtained from Sveriges Riksbank.
- **Inflation** (π_t): year-on-year CPIF inflation (consumer price index with fixed interest rate), sourced from Statistics Sweden (SCB). CPIF is the Riksbank's formal target variable.
- **Inflation expectations** ($\exp_t$): one-year-ahead CPI inflation expectations from the Prospera (now Origo) survey of money market participants. The monthly values were manually extracted from the original PDF survey reports by the author, covering January 2010 to December 2024.[1]
- **Slack** (gap_t): the seasonally adjusted unemployment rate, obtained from SCB. In the baseline specification, slack is proxied by the unemployment rate rather than a detrended unemployment or output gap. This follows standard practice in monthly policy-rule estimation, where real-time output-gap measures are unreliable and unemployment gaps are highly sensitive to filtering (Orphanides and van Norden, 2002). Because all regressors enter the TVP system in standardized form (see Methodology), the coefficients capture the evolving response to relative movements in unemployment rather than its level. Using the unemployment rate as a high-frequency slack proxy is common in empirical Taylor-rule studies and avoids noise from monthly detrending.

[1] The Prospera money market panel is the Riksbank's primary gauge of professional inflation expectations and is routinely used in monetary policy analysis. Although the survey refers to CPI rather than CPIF, the difference is minor at the one-year horizon, and the expectations target level has always been 2 per cent.

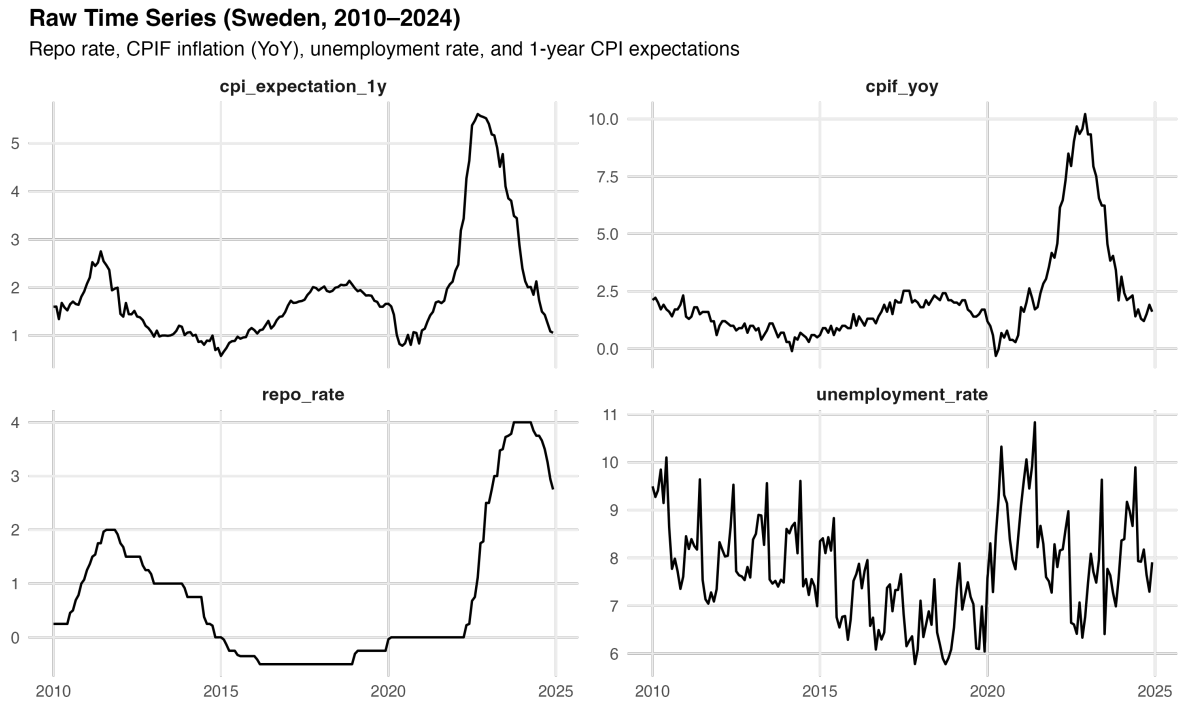


Figure 1: Repo rate, CPIF inflation, inflation expectations and unemployment, 2010–2024.

A practical issue concerning the choice of expectations measure would be that the Riksbank's inflation target is defined in terms of CPIF. However, the longest and most consistent monthly expectations series refers to one-year-ahead CPI inflation from Prospera's money market survey. This series is the Riksbank's primary measure of professional expectations and is widely used in policy analysis (Sveriges Riksbank, 2017, 2021). CPIF differs from CPI mainly by excluding mortgage interest expenditure. The two series track each other closely at the one-year horizon, except during periods of sharp mortgage-rate movements (Jonsson and Österholm, 2009). Following standard Swedish practice, I therefore use CPI-based expectations as a proxy for general inflation expectations around the 2 per cent target. CPIF expectations are only available for recent years and at lower frequency, whereas the CPI-based Prospera series provides a long, internally consistent monthly panel. Since the model is estimated using deviations from the target and CPI and CPIF inflation co-move strongly, any measurement differences are unlikely to affect the qualitative time variation in the estimated coefficients (Flodén, 2012). This choice is nevertheless noted as a limitation.

ADF and KPSS tests indicate that the repo rate, inflation gap, and expectations gap are highly persistent and behave as non-stationary processes, while the unemployment-based slack measure appears stationary. This pattern is typical in macroeconomic data and consistent with earlier work on time-varying policy rules (Cogley and Sargent, 2001b, 2005b; Primiceri, 2005). The state-space TVP model however, does not require stationary regressors, where the Kalman filter handles persistent or near-integrated inputs without affecting identification. Standardizing the regressors prior to estimation is therefore a numerical stabilization step rather than an attempt to enforce stationarity. Descriptive statistics show that inflation and one-year expectations move very closely together (correlation around 0.9), while unemployment is only weakly related to inflation and the policy rate. This suggests that slack plays a secondary role relative to prices and expectations in the monthly policy rule.

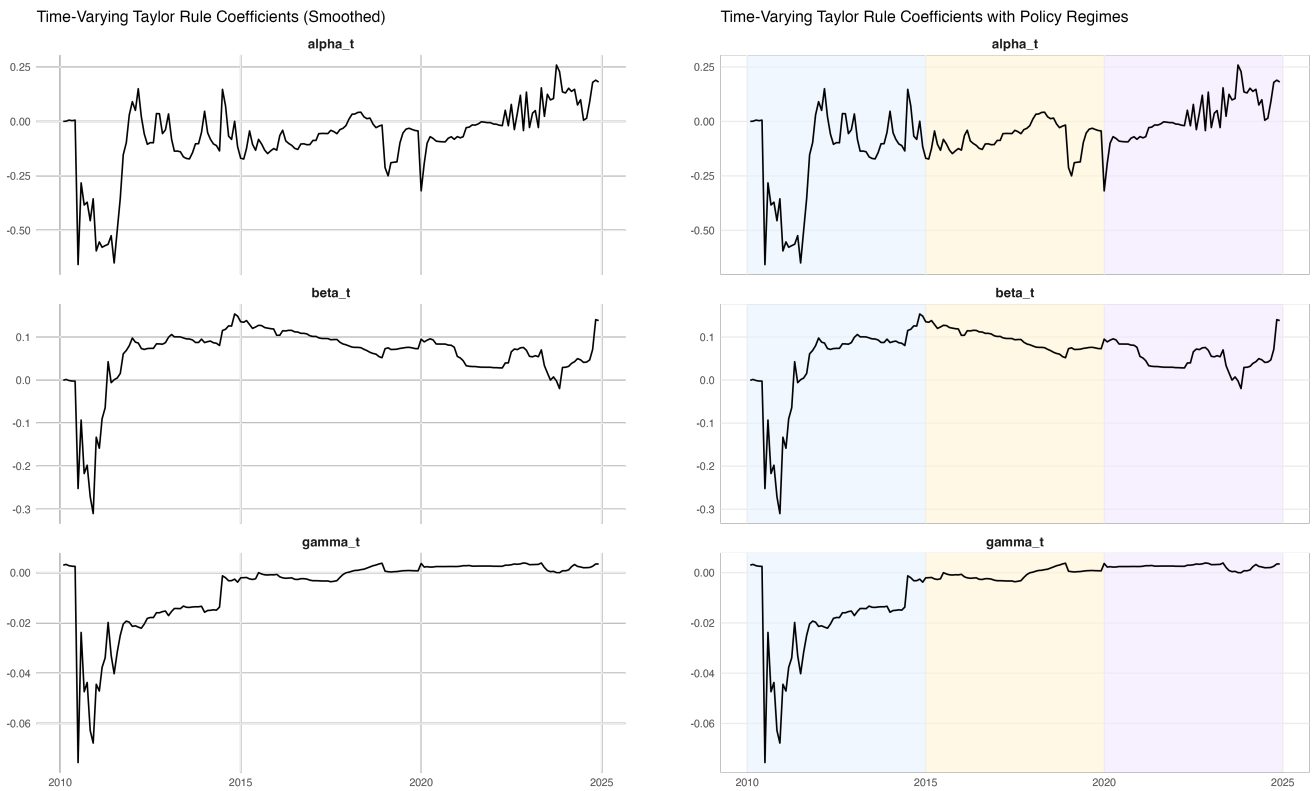
Results

Figure 2a plots the smoothed time paths of the three Taylor-rule coefficients. The response to realized inflation, α_t, exhibits three distinct phases. First, it falls sharply in 2010, reaching its most negative value in the entire sample, before rebounding quickly toward zero. Then, during roughly 2012–2020, α_t

oscillates around zero and crosses into positive territory several times, but without establishing a sustained positive pattern. Finally, from late 2021 onward, α_t rises more decisively and remains persistently above zero, indicating a structural strengthening of the Riksbank's direct response to realized inflation during the post-pandemic inflation surge. The expectations coefficient, β_t, is initially small and briefly turns clearly negative in 2010–2011, reaching its global minimum at the end of 2010. It then rises sharply into positive territory by 2012 and remains positive for most of the sample, with a gradual decline after the mid-2010s and a temporary softening around the pandemic. The slack coefficient, γ_t, is strongly negative at the beginning of the sample but rises quickly toward zero and then remains very close to zero for the rest of the period, indicating only a modest and diminishing role for the unemployment rate in the policy rule.

Figure 2b shows the same coefficient paths with shaded bands corresponding to three policy regimes: (i) the post-financial-crisis, low-inflation period (2010–2014); (ii) the negative-rate and QE period (2015–2019); and (iii) the pandemic and inflation-surge period (2020–2024).[2] Although the precise dating is approximate, the qualitative patterns in the estimated coefficients are robust to reasonable alternative cutoffs. The regime shading highlights that the strongest negative values of α_t and γ_t occur early in the sample, when inflation was persistently below target and slack remained elevated, whereas the response to expectations is most clearly positive during the second regime.

Table 1 summarizes the regime-average coefficients. On average, α_t is negative in the first two regimes (around -0.16 and -0.08, respectively) and turns mildly positive in 2020–2024. By contrast, β_t is positive throughout, with a regime-average of about 0.09 during the negative-rate and QE period and around 0.05 in the most recent regime. The slack coefficient γ_t becomes less negative over time and is close to zero from 2015 and onward. Overall, these patterns suggest that the Riksbank's rule placed a more persistent weight on inflation expectations than on realized inflation or unemployment over most of the sample.



(a) Smoothed TVP coefficients, 2010–2024. **(b)** TVP coefficients with monetary policy regimes.

Figure 2: Time-varying Taylor rule coefficients: smoothed estimates (left) and regime-shaded representation (right).

[2]The regime boundaries follow widely used interpretations of Swedish monetary policy. The introduction of negative interest rates and quantitative easing in 2015 marked a clear break from the preceding low-inflation period, as documented in the external review of Riksbank policy (Goodfriend and King, 2016a). Likewise, the COVID-19 shock in early 2020 and the subsequent inflation surge are widely treated as a distinct macroeconomic regime.

Table 1: Regime-Average TVP Coefficients, 2010–2024

Regime	Obs	α_t mean	β_t mean	γ_t mean
2010–2014: Post-crisis, low inflation	59	−0.161	0.033	−0.020
2015–2019: Negative rates & QE	60	−0.076	0.094	−0.001
2020–2024: Pandemic & inflation surge	60	0.019	0.053	0.002

Given the paper's focus on the expectations–policy relationship, the dynamics of β_t warrant a closer look. Figure 3 plots only the expectations coefficient, again with shaded regimes, and marks the dates at which β_t attains its global minimum and maximum. The coefficient falls sharply in 2010, with a global minimum of about −0.31 in December 2010. Then increases steadily to a global maximum of roughly 0.15 in November 2014 (see Table 2). After 2014, β_t remains strictly positive for several years but drifts gradually downward during the negative-rate and QE period. The coefficient then exhibits a temporary decline around the onset of the pandemic, though it stays above zero throughout this phase, before rising again during the initial inflation surge of 2021–2022. A more pronounced and short-lived drop occurs in mid-2023, during which β_t briefly turns negative, after which the coefficient recovers sharply and ends the sample near its earlier highs. Taken together, these dynamics indicate that while the expectations response weakened at times, it remained an important and generally positive component of the policy rule over most of the sample, with only brief interruptions. The early negative values suggest that, during the immediate post-financial-crisis years, higher inflation expectations were sometimes associated with a more cautious or even counter-intuitive policy response. By the mid-2010s, however, the rule had shifted decisively toward a positive expectations response. When survey expectations rose relative to target, the estimated rule prescribes a higher policy rate, and vice versa. This positive responsiveness is especially pronounced during the negative-rate and QE regime, where β_t remains consistently above zero. Although a brief negative dip appears in 2023, the coefficient quickly recovers and ends the sample at elevated levels, reinforcing the view that inflation expectations served as a meaningful and generally positive policy input throughout most of the period.

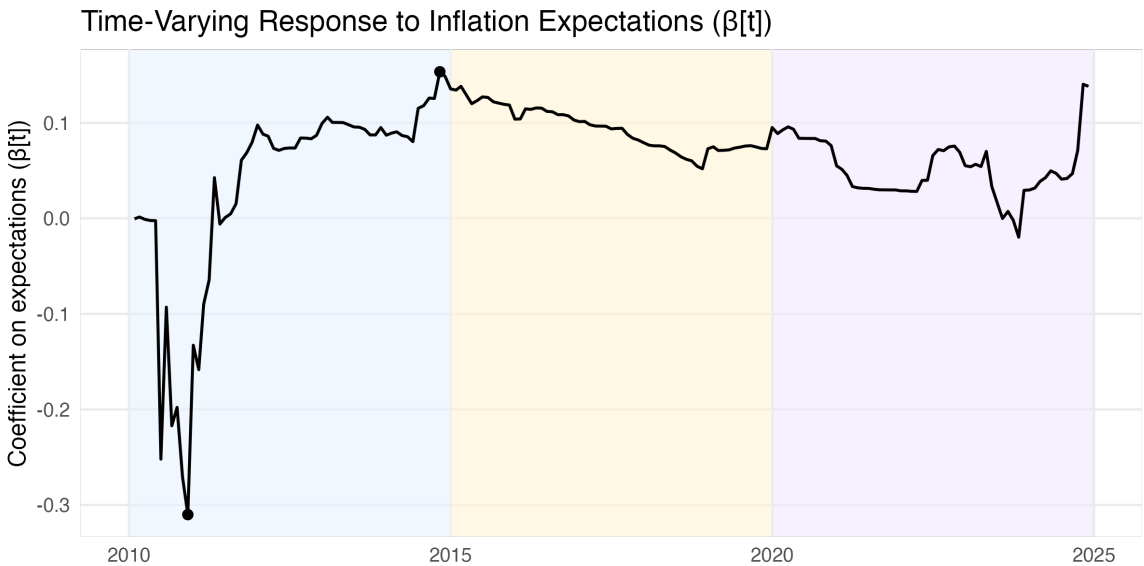


Figure 3: Time-Varying Response to Inflation Expectations (β_t). Shaded areas denote the same three monetary policy regimes as in Figure 2b. Dots mark the global minimum and maximum of β_t.

Table 2: Global Minimum and Maximum of the Expectations Coefficient β_t

Date	β_t	Regime
2010-12-01	-0.310	2010–2014: Post-crisis, low inflation
2014-11-01	0.153	2010–2014: Post-crisis, low inflation

To evaluate how well the time-varying rule explains actual policy decisions, Figure 4 compares the observed repo rate with the rate implied by the estimated TVP Taylor rule, where the fitted path tracks the broad movements of the policy rate, including the gradual reductions into negative territory, the prolonged low-rate period, and the sharp tightening during 2022–2023. Some short-run deviations remain, especially around episodes where policy decisions may have reflected financial stability concerns or communication strategies not captured by the simple rule.

Table 3 summarizes the in-sample fit. The root mean squared error (RMSE) is close to 4 basis points, and the mean absolute error (MAE) is roughly 2.5 basis points. These small residuals suggest that the combination of interest-rate smoothing, inflation and expectations gaps, and slack captures a substantial fraction of the variation in the monthly policy rate over 2010–2024. At the same time, the TVP specification is flexible by construction, so these fit measures should be interpreted primarily as evidence that the rule is internally coherent rather than as proof that policy strictly followed this exact reaction function. The estimated interest-rate smoothing parameter is high, (0.98), and the state variances imply that the inflation coefficient is allowed to drift more over time than the expectations and slack coefficients, which remain relatively smooth.

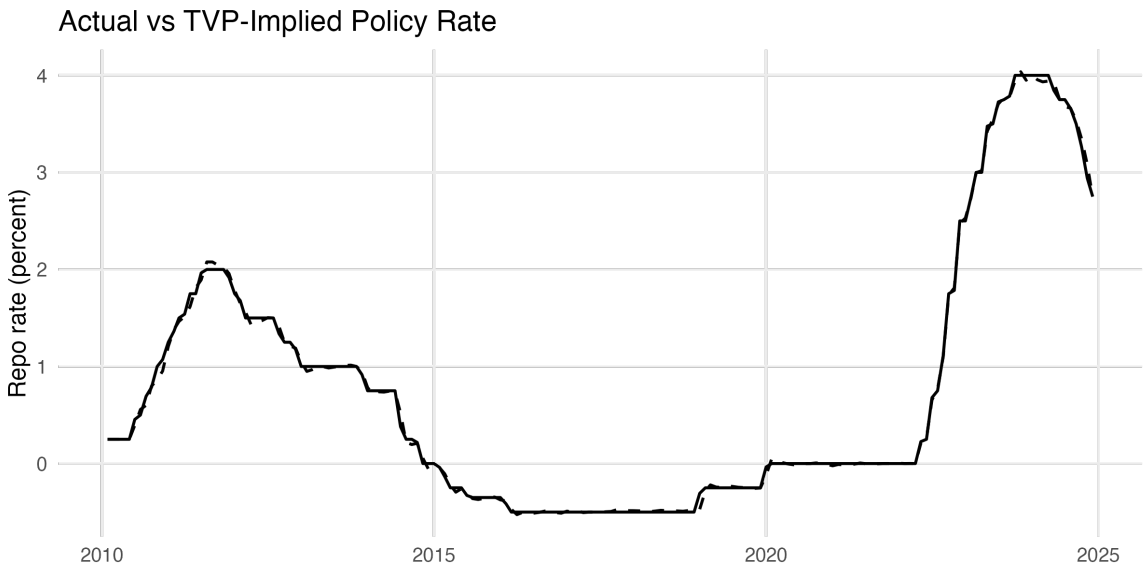


Figure 4: Actual vs TVP-Implied Policy Rate. The dashed line shows the policy rate implied by the estimated time-varying Taylor rule; the solid line is the observed repo rate.

Table 3: In-Sample Fit of the TVP Taylor Rule

Statistic	Value
RMSE	0.0397
MAE	0.0255

Discussion

The empirical results reveal several important features of how Swedish monetary policy evolved over 2010–2024. First, the Riksbank's response to inflation expectations, captured by β_t, is consistently positive after 2012 and essentially remains so across all three monetary policy regimes. This suggests that expectations played a stable and meaningful role in the policy rule, even when the policy rate was constrained by the effective lower bound. This finding aligns with the strong emphasis the Riksbank places on expectations in its communication and with evidence that expectations help predict policy decisions (Jonsson and Österholm, 2009; Lundmark, 2023). The pattern of β_t also sheds light on regime-specific policy behavior. The sharp rise in 2012–2014 coincides with rising concern about inflation being persistently below target.

During this period, the Riksbank faced criticism for keeping rates too high, and both internal minutes and external evaluations highlight increased focus on expectations anchoring (Svensson, 2014; Goodfriend and King, 2016b; Sveriges Riksbank, 2013). The temporary negative values in 2010–2011 are better viewed as a mismatch between expectations dynamics and gradual policy adjustments in the turbulent post-financial-crisis environment rather than a structural negative response. After 2015, β_t declines somewhat but remains positive, indicating that expectations continued to matter even during negative rates and QE, although with less weight than during the low-inflation years. By contrast, the response to realized inflation, α_t, shows a more complex pattern. The coefficient plunges into negative territory in 2010–2011, then fluctuates around zero for much of the decade. Between 2012 and 2020 it exhibits several short-lived positive episodes but generally remains weak, consistent with a policy environment in which backward-looking inflation played a limited role, while forward-looking indicators, including expectations, were more prominent. A clearer shift occurs only after 2021, when α_t rises decisively and remains positive throughout the inflation surge. This upward movement aligns with the global tightening cycle in which central banks, including the Riksbank, responded more forcefully to realized inflation as it became the dominant macroeconomic concern. The slack coefficient γ_t follows a more predictable trajectory. Its large negative values in the early 2010s reflect post-crisis labour market conditions, while the subsequent convergence toward zero indicates a diminishing role for slack in policy decisions. This is consistent with Riksbank communications during the recovery (Riksbank, 2011, 2013). After 2015, γ_t hovers near zero, in line with evidence of a flattening Phillips curve in Sweden and other advanced economies (Bank for International Settlements, 2019), which reduces the usefulness of slack in high-frequency policy setting. The fitted-values analysis shows that the TVP rule tracks the actual policy rate closely, with residuals rarely exceeding a few basis points. This suggests that a small number of evolving coefficients, combined with interest-rate smoothing, explains much of the monthly variation in the policy rate. However, the close fit should be interpreted with caution: a TVP model is flexible by construction, and part of the drift in coefficients may reflect endogenous adjustments rather than a structural rule. Episodes where the fitted and actual rates diverge, such as during the pandemic or periods of financial instability, suggest that additional factors not included in the rule sometimes influenced policy decisions. A key methodological limitation concerns the treatment of the long-run neutral interest rate. Because the TVP rule is estimated in levels rather than in policy-rate gaps ($i_t - i_t^*$), slow-moving movements in the natural rate may be absorbed mechanically by drifting coefficients, particularly α_t and β_t. This implies that some estimated time variation may reflect changes in i_t^* rather than shifts in policy behavior. Ideally, a longer sample and external estimates of the natural rate (e.g. Laubach–Williams) would help distinguish equilibrium-rate movements from genuine adjustments in the reaction function. Overall, the results indicate that Swedish monetary policy gradually shifted from a heavier reliance on expectations (and to a lesser extent slack) in the low-inflation early 2010s toward a more balanced rule in which the response to realized inflation strengthened after 2021. The time-varying approach thus uncovers behavioral adjustments that constant-parameter Taylor rules would miss and adds nuance to the interpretation of Swedish policy over a period marked by significant structural change.

Conclusion

This paper has analyzed how the Riksbank's monetary policy rule evolved over 2010–2024 by estimating a time-varying Taylor-type reaction function in state-space form. Allowing the coefficients to drift over time reveals systematic changes in the policy response to realized inflation, inflation expectations, and economic slack. These are changes that would remain obscured in a constant-parameter framework. The results shows that the Riksbank consistently placed an almost perfectly positive weight on inflation expectations after 2012, while on the other hand the response to realized inflation fluctuated around zero for much of the sample and only became more persistently positive during the post-2021 inflation surge. The response to expectations, captured by β_t, is particularly important for understanding Swedish monetary policy during this period. It rises sharply in the early 2010s as concerns about a "too-low" inflation intensified, reaches its maximum just before the negative-rate and QE regime, while after that

remaining positive throughout nearly the entire sample. Regime averages further show that the QE period is associated with the strongest expectations response, consistent with the Riksbank's emphasis on expectations anchoring when conventional policy space was constrained. Although β_t briefly dips during the pandemic and again in 2023, it quickly recovers and ends the sample at elevated levels, indicating that expectations remained an crucial component of the effective policy rule. By contrast, the inflation and slack coefficients exhibit weaker and more muted patterns. The inflation coefficient α_t is negative or close to zero for much of the 2010s, becomes temporarily positive at several points, and turns more durably positive only once inflation accelerated after 2021. The slack coefficient γ_t is strongly negative in the early post-crisis period but converges steadily toward zero after 2013, remaining near zero thereafter. This notably depreciating role for slack fit the picture from broader evidence of a flattening Phillips curve in Sweden, limiting information content of unemployment gaps for high-frequency policy adjustments. While the estimated time-varying rule captures the broader path of the policy rate and provides a coherent narrative of evolving policy behavior, the analysis also points to its limitations. Since a TVP model is inherently flexible, and there are occasional divergences between fitted and actual policy rates, I would suggest that factors such as concerns about financial stability, exchange-rate considerations, or communication objectives occasionally influenced policy decisions beyond the scope of the simple rule. Moreover, because the model is estimated in levels rather than as a policy-rate gap, part of the slow-moving trend in the neutral rate may be absorbed by the drifting coefficients, particularly α_t and β_t. Future work could incorporate stochastic volatility, alternative expectations measures, or a multivariate TVP-VAR structure, or explicitly model movements in the neutral rate to distinguish cyclical adjustments from trends in the policy stance. Nonetheless, the results still demonstrates that a time-varying approach do provide valuable insight into how the expectations policy relationship in Sweden evolved through periods of extended low inflation, unconventional policy measures, and the sharp inflation resurgence of the early 2020s.

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